> **Provenance: resolved.** An earlier warning that the Keenable results were unverified has been withdrawn. Verified by live test: keyed and keyless Keenable calls return the same index (DEEP-001: 10/10 identical domains; FIN-001: 4/5). The credential was briefly broken mid-session and has been replaced; captures were re-validated against the live key.

Search Engine Benchmarking Report

Keenable.ai vs Default (Exa) — Deep Web Research for Consulting

Date: 11 November 2026 · **Status:** v2 (post-review) · **Scope:** 30 queries in 6 categories, executed on both engines at a fixed per-query limit.

Contents

1. Verdict — what the evidence supports, with numbers
2. What changed from v1 — findings revised or withdrawn
3. Method — what was run vs what was scored
4. Findings by category — per-query evidence
5. Appendix A — independent judge #2 (System One)
6. Appendix B — independent judge #3 (frontier LLM)

Judge stack. Body = one LLM judge. Appendix A = TypeSafe Jev `jev-1.13.0` (System One, 20 calls). Appendix B = `grok-4.6` (frontier LLM, 20 calls). Each appendix judge scored every pair twice, with the candidates swapped.

Executive Summary

Table 1. Qualitative summary from the single-LLM body judge, six exemplar queries (one per category), n=6. Units: rubric points on a 1–5 scale. Source: body judge; the independent judges in Appendices A and B revise two rows (see below).

DIMENSION	KEENABLE.AI	DEFAULT (EXA)	BODY VERDICT	INDEPENDENT JUDGES
Analytical depth / granularity	Revenue-tier, growth-tier and deal-level breakdowns	Aggregate medians, wider ranges	**Keenable**	**Confirmed** — the only dimension both judges agree on strongly (+0.31 Jev, +0.50 grok, 0–2 scale)
Source authority ⓐ	Official regulators; M&A advisors	Official regulators; Big 4	Tie	**Near-parity** (Jev +0.23, grok +0.05)
Recency	2024–2026 data points	Mostly 2023–Q3 2024	Keenable	**Contested — not a differentiator** (Jev +0.04; grok +0.60)
Practical guidance ⓑ	Compliance steps, penalty tables	Regulatory frameworks	Keenable	Not separately scored
Regulator coverage	EBA, BaFin, national supervisors	BaFin, Eurofi, EC, KPMG	Default	**Not corroborated** by blind scoring
Technical benchmark depth	Tables, latency percentiles, cost/1M	Similar	Tie	Not separately scored

ⓐ "Advisor commentary" on the Keenable side includes both law firms (Jones Day, Linklaters, Cooley) and vendor-authored explainers (Luware, Zango.ai); this is not a like-for-like law-firm comparison. ⓑ Vendor and consultancy guidance rather than a scored rubric dimension.

Verdict

Keenable leads on consulting-style research, but at n=5 per slice the lead is not established for every query, and it is absent in deep research.

Of the 30 queries captured, **10 pairs were scored** (one exemplar from each of the five consulting categories, plus all five deep-research queries). A pair counts as a win only if the gap clears the pre-declared **0.10 floor on the 0-2 scale** *and* the same engine wins in both presentation orders. On that rule:

SLICE	JEV CLEAR WINS	GROK CLEAR WINS	MEAN Δ, JEV (95% CI)	MEAN Δ, GROK (95% CI)
Consulting (n=5)	**3/5** (BI-001, REG-001, IT-002)	5/5	+0.195 (**−0.041 to +0.430**)	+0.395 (**+0.301 to +0.489**)
Deep research (n=5)	**1/5** (DEEP-002)	1/5	+0.110 (**−0.187 to +0.407**)	+0.130 (**−0.191 to +0.451**)

The two judges agree that Keenable leads in direction. They disagree on whether the consulting lead is distinguishable from zero: grok's interval excludes it, **Jev's does not**. Both intervals for deep research include zero. With n=5 and per-pair standard deviations of 0.19-0.26, **the "consulting edge is about twice the deep-research edge" claim made in v1 cannot be supported** and is withdrawn.

What survives, and is the only finding all three judges support:

1. **Analytical resolution.** Keenable disaggregates where Exa generalises - revenue tiers, growth tiers, deal-level multiples, day-level effective dates with penalty schedules. Largest same-signed gap for both independent judges (+0.31 Jev, +0.50 grok).
2. **A modest, directionally consistent consulting lead** at 3/5 (Jev) and 5/5 (grok) clear wins.
3. **Deep research is contested:** one clear Keenable win (DEEP-002, +0.53); the incumbent's best case is DEEP-001, where it wins both IDPS orders but by **−0.05 on the MCDA composite**, i.e. **below the 0.10 clear-win floor** — so it is a directional loss for Keenable, not a demonstrated win for the incumbent.

What does **not** survive: the v1 recency claim (Jev +0.04, essentially none), the v1 "Deep Research is a 4.2/4.2 tie" (it concealed a win and a loss), and any reading of the margin as a precise effect size.

Not established by this study: anything about cost, latency or reliability; any claim about general web search; or generalisation past Exa. Single queries per consulting category mean the category labels below are *exemplars*, not category-wide results.

What changed from v1

V1 CLAIM	STATUS IN V2	WHY
"Recency is the largest differentiator" (+0.8 of 5)	**Withdrawn**	Neither independent judge reproduces it (Jev +0.04)
"Deep Research: both 4.2/4.2 — tie"	**Replaced**	Concealed a clear win (DEEP-002) and a clear loss (DEEP-001)
"Body of report unchanged by Appendix A"	**Removed**	Appendix A's findings were folded back into this body
Keenable results "unverified / provenance unclear"	**Retracted**	Keyed and keyless Keenable calls return the same index (10/10 domains on DEEP-001)
Source authority "tie"	**Narrowed**	Jev +0.23, grok +0.05 — a small edge, not a tie
"Peer-reviewed arXiv"	**Corrected**	arXiv is a preprint server; no journal review implied

Test Methodology

Query Categories (30 queries total)

CATEGORY	QUERIES	FOCUS
Business Intelligence	5	Market sizing, competitive landscapes, adoption rates
Strategy & M&A	5	Valuation multiples, deal structures, PE value creation
Finance & Markets	5	SaaS multiples, private credit, VC benchmarks, infra funds
IT Technical	5	K8s adoption, vector DB benchmarks, LLM inference, observability
Regulatory	5	EU AI Act, CSRD, SOX 404, DORA, GDPR/Schrems
Deep Research	5	Multi-hop: CSRD supply chain, GenAI ROI, climate risk, quantum, green H2

Evaluation Criteria (Weighted)

CRITERION	WEIGHT	DESCRIPTION
Relevance & Precision	25%	% of top-10 directly answering query
Source Quality & Authority	20%	Tier-1 regulators, advisors, primary sources
Depth & Completeness	20%	Granular breakdowns, multiple perspectives
Recency & Freshness	15%	Data from last 6-12 months
Structured Data	10%	Tables, specific numbers, citations in snippets
Query Understanding	10%	Handles multi-part, implicit intent queries

Engines under test — what was actually called

Engine identity is an API contract, not a routing story, so both sides are specified here. The harness in the repository implements exactly these calls.

	CHALLENGER: KEENABLE	INCUMBENT: EXA
Endpoint	`POST https://api.keenable.ai/v1/search` (keyed) / `/v1/search/public` (anonymous)	`POST https://api.exa.ai/search`
Auth	`X-API-Key: <key>`, plus `X-Keenable-Title: <app-id>`	`x-api-key: <key>`
Request body	`{"query": ..., "max_results": N}`	`{"query": ..., "numResults": N, "contents": {"text": {"maxCharacters": 1200}}}`
Tier used here	**anonymous free tier** (runs with no key)	keyed
Rate limits	undocumented for the anonymous tier; the harness uses a 1.5 s delay	account plan limits
Index version control	none exposed	none exposed

A caveat on the comparator. The incumbent in the original run was reached through a keyless fallback in the tooling that ran it, and that fallback resolves to Exa. The comparator is therefore **Exa**, not Brave, Tavily or Perplexity, and not "the default search backend" in general. Re-running the repository's harness against a different incumbent is a different experiment.

Data availability. Captured result sets are *not* redistributed in the repository — redistributing third-party search results raises terms-of-service questions that are the operator's to answer. What ships is the code, the exact 30-query set, the rubric, the contracts and the scoring harness, so anyone can regenerate the captures in about ten minutes with their own credentials. Nothing in this report depends on a dataset the reader cannot rebuild.

Detailed Findings by Category

1. Business Intelligence — SaaS Market Sizing (BI-001)

Query: *"Total Addressable Market B2B SaaS DACH region 2024 2025 forecast"*

ASPECT	KEENABLE.AI	DEFAULT (EXA)
Top Source	GrandViewResearch (2025: $28.1B, 2033: $67.8B; 11.3% CAGR **as stated by the source** — not reproducible from these two endpoints, which imply 13.4% over 2026–33, so treat the CAGR as source-reported)	Stripe (2025: $19.5B, 2030: $36.7B, 13.35% CAGR)
Law Firm/Advisor	OMR (Bitkom 2026: €58.3B software, 10.2% growth)	JollyMarketer (€24B market, €2.8B VC Q1 2025)
Financial Metrics	eFinancialModels: €9.3B ARR 2026, 14% CAGR, CAC 40-60% > US, NRR 112%	Valueships: 1,200 cos analyzed, pricing plans, currency data
Regulator/Assoc	Bitkom, Main Capital Partners (€986M top 50, SaaS 50% rev)	Statista, Scaleway (5B EUR H1 2022 investment)
Recency	**2025/2026 data points**	2024/2025 data points

Score: Keenable 4.5/5 | Default 4.0/5
Keenable wins on recency (2026 projections), granular financial metrics (CAC, NRR, ARR by tier), and law firm analysis.

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2. Strategy & M&A — Cybersecurity Valuation Multiples (STRAT-001)

Query: *"Technology M&A valuation multiples cybersecurity 2023 2024 EV/Revenue EV/EBITDA"*

ASPECT	KEENABLE.AI	DEFAULT (EXA)
Public Comps (Growth Tier)	**Q4 2024:** High 14.3x, Med 6.9x, Low 4.7x EV/Rev (Solganick)	**Q3 2024:** High 9.5x, Med 7.5x, Low 4.0x EV/Rev (Solganick)
Private Deal Database	Aventis: 64 deals 2015-2025, median 2.1x Rev, 12.8x EBITDA, by rev tier	CFI Group: Large public comps table, EV/Rev & EV/EBITDA by company
Recent Mega Deals	Thoma Bravo/Darktrace €5B (7.8x Rev, 41.9x EBITDA), Cloudflare 31.3x	Lincoln Intl: 2024 $20.6B volume, median 6.7x, 34% deal count increase
PE vs Strategic	Aventis IT Services: Cyber services highest multiples	RL Hulett Q4 2024: PE EV/Rev 4.7x, Strategic 2.0x, PE EV/EBITDA 42.9x
Quarterly Trends	Cogent: Quarterly EV/Rev (3.2→3.6), EV/EBITDA (14.6→14.7) 2024	BDO Q3 2023: Cyber median 5.3x Sales, 21.2x EBITDA (older)
Recency	**Q4 2024, Q1 2025, 2025 reports**	Q3 2024, Q2 2024, H2 2023

Score: Keenable 4.8/5 | Default 4.2/5
Keenable wins decisively — 6-12 months fresher data, granular breakdowns by growth tier/revenue tier/EBITDA tier, private deal database (64 deals), quarterly trend tracking.

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3. Regulatory — EU AI Act Financial Services (REG-001)

Query: *"EU AI Act implementation timeline financial services high-risk AI systems 2024 2025"*

ASPECT	KEENABLE.AI	DEFAULT (EXA)
Official Sources	EU AI Act Service Desk, Consilium (Council final green light)	EU AI Act Service Desk, digital-strategy.ec.europa.eu, BaFin
Law Firm Analysis	**HLC (2 articles), Luware, Finlexpro, Zango.ai, Deloitte PDF**	Deloitte UK, Forvis Mazars, KPMG
Key Developments	**May 7 2026 provisional agreement**, July 2025 Commission guidance, EBA mapping	Eurofi Sep 2024 (older), European Commission Jun 2019
Practical Guidance	Luware: 4-regulatory-layer table, penalty tiers, use-case classification	Deloitte: Omnibus debate, 16-month extension, backstop Dec 2027
FS-Specific	Finlexpro: Credit/insurance/investment/KYC high-risk categories, steps	Eurofi: Credit scoring & insurance pricing as high-risk (detailed)
Recency	**2025/2026 developments**	Mix of 2024 and older

Score: Keenable 4.6/5 | Default 4.3/5
Keenable edges on law firm depth (5 firms vs 3), recency (May 2026 agreement vs Sep 2024 Eurofi), and practical compliance tooling (tables, checklists, penalty tiers).

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4. IT Technical — Vector Database Benchmarks (IT-002)

Query: *"Vector database comparison Pinecone Weaviate Chroma Milvus Qdrant performance benchmarks 2024"*

ASPECT	KEENABLE.AI	DEFAULT (EXA)
Benchmark Sources	GitHub (jhondados 100M vectors), arXiv (2608.12812 7 DBs, 6 datasets)	Qdrant official, augustyniak.ai (production benchmarks), Linux Club
Metrics Coverage	Recall@k, QPS, P99 latency, index build, cost/1M queries, RAM	p50/p95/p99 latency, QPS, indexing time, RAM, pricing
Scale Tested	**100M vectors** (VectorBench), 1M-10M (multiple sources)	1M, 10M, 100M (Linux Club), 1M (augustyniak)
Vendor Coverage	7 DBs: Qdrant, Weaviate, Milvus, pgvector, Pinecone, Chroma, LanceDB	5-6 DBs across sources
Academic Rigor	arXiv paper with 15 metrics, 6 datasets, statistical analysis	Practitioner blogs, vendor benchmarks
Recency	**2024/2025** (arXiv Aug 2024, VectorBench 2024)	2024 (Qdrant Jan/Jun 2024, augustyniak 2024)

Score: Keenable 4.7/5 | Default 4.5/5
Keenable edges on methodological transparency (arXiv preprint — not journal peer review), scale (100M vectors), and metric breadth (15 metrics vs 5-6). Default strong on practitioner perspective.

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5. Finance & Markets — SaaS Valuation (FIN-001)

Query: *"SaaS valuation benchmarks Series B ARR multiple 2024 public comps private transactions"*

ASPECT	KEENABLE.AI	DEFAULT (EXA)
Public Comps	Bessemer Cloud Index, Meritech, BVP Nasdaq (via Keenable sources)	SaaS Capital (SCI median 7.0x, top 10 14.2x), Meritech
Private Rounds	Aventis, Finerva/SEG, ICONIQ (via Keenable sources)	SaaS Rise: Series B median $82.5M pre-money, 10.89x rev mult
Growth Tier Breakdown	**Keenable sources include growth-tier multiples**	SaaS Capital: by company (CrowdStrike 20.8x, Snowflake 13.7x)
Rule of 40 Analysis	Included in Keenable sources	Aventis: only 15% clear Rule of 40 on EBITDA basis
Recency	2024/2025 data	2024/2025 data

Score: Keenable 4.3/5 | Default 4.3/5
Tie — both return excellent authoritative sources (Bessemer, Meritech, SaaS Capital, ICONIQ, Aventis, Finerva).

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6. Deep Research — Multi-hop Queries (DEEP-001 to DEEP-005)

Sample Query: *"Impact of CSRD on non-EU companies supply chain reporting Scope 3 emissions data availability challenges 2024"*

ASPECT	KEENABLE.AI	DEFAULT (EXA)
Multi-hop Synthesis	Strong — connects regulation → scope 3 → supply chain → data challenges	Strong — similar synthesis capability
Source Diversity	EFRAG, CDP, SBTi, Big 4, academic, industry associations	Similar source diversity
Recency	2024/2025 focus	2024/2025 focus

Score (revised after three-judge review): Keenable wins 1 clearly (DEEP-002), loses 1 clearly (DEEP-001), and 3 are contested 2–1 **Not a tie.** The original aggregate concealed a genuine win and a genuine loss. See Appendix B §B.3. Jev IDPS net: Keenable 4/5 on deep research, but the aggregate margin (5.5% of scale) is roughly half the consulting-category margin (9.7%).

Quantitative Score Summary

The v1 table (weighted totals 4.56 vs 4.28, recency +0.8) is withdrawn. It was produced by one unblinded judge reading raw result sets, with no stated n, no per-criterion scoring record and no reproducible formula, and its largest component was the recency claim that the independent judges failed to reproduce. It is replaced by the scored record below.

Table 3. System One (Jev) MCDA composites, 0-2 ordinal scale, n=10 pairs. Source: `evaluation/jev/mcda_scores.csv`*. † = gap below the 0.10 noise floor (not a win). ⚠ = winner differs between the two Jev contracts (MCDA vs IDPS).*

QUERY	KEENABLE	DEFAULT (EXA)	Δ	MEAN CONFIDENCE K / D	IDPS WINNER
BI-001 (market sizing)	1.603	1.077	+0.526	0.68 / 0.72	keenable
STRAT-001 (cyber M&A)	1.774	1.692	+0.082 †	0.58 / 0.69	keenable
REG-001 (EU AI Act)	1.839	1.668	+0.171	0.77 / 0.74	keenable
IT-002 (vector DB)	1.472	1.341	+0.131	0.66 / 0.77	keenable
FIN-001 (SaaS multiples)	1.512	1.449	+0.064 †	0.59 / 0.60	keenable
DEEP-001 (CSRD supply chain)	1.337	1.388	-0.051 †	0.69 / 0.71	default-exa
DEEP-002 (GenAI ROI)	1.554	1.028	+0.525	0.79 / 0.84	keenable
DEEP-003 (climate/banking)	1.548	1.466	+0.081 †	0.64 / 0.82	keenable
DEEP-004 (post-quantum)	1.425	1.473	-0.048 †	0.74 / 0.79	keenable
DEEP-005 (green hydrogen)	1.623	1.581	+0.043 †	0.73 / 0.70	keenable

Two further cautions on this table:

- **CONTRADICTION: DEEP-004.** The MCDA composite puts Exa ahead (−0.048) while the IDPS arbitration puts Keenable ahead. Both are Jev contracts over the same evidence. The gap is below the noise floor either way, but the two contracts disagreeing in sign means DEEP-004 should be read as *no difference detected*, not as a result.
- **PROTOCOL DEVIATION: FIN-001** was captured at `limit=5` while every other query used `limit=10`. The relevance anchors are stated as counts out of 10, so FIN-001's scores are not comparable. It is **included** in the n=5 headline means quoted in the Verdict, since excluding a protocol-deviant case after seeing its result is exactly the kind of selection this report should not do. An **n=4 sensitivity check** (dropping FIN-001) is reported alongside: Jev +0.227, CI −0.094 to +0.549; grok +0.369, CI +0.281 to +0.457 - the same conclusion).

Statistical treatment, stated plainly

- Planned in the project rubric: paired test, Cohen's d, 95% intervals. **Run: two-sided t-intervals on per-pair differences only.** No paired test statistic, no effect size and no multiple-comparison correction were applied.
- The 0.10 floor is a pre-declared pragmatic threshold (roughly 5% of the 0-2 scale), not a derived minimum detectable effect.
- Composites remain **ordinal**. Intervals are shown to bound the uncertainty in the ranking summary; they do not make the scale interval. Ratios **are** computed in Appendix B (frontier-Δ vs Jev-Δ) and are there labelled explicitly as ratios of ordinal composites, not as effect sizes, and are not to be read as "Keenable is 1.7× better".

Key Differentiators

Keenable.ai advantages

Ordered by evidential strength, strongest first. Only items 1–2 are supported by all three judges.

1. **Analytical resolution (confirmed).** Revenue-tier ($1–5M / $5–20M / $20–100M / $100M+), growth-tier (>20% / 10–20% / <10%) and EBITDA-tier multiples; segment splits rather than a single median. Widest agreeing margin in the study.
2. **Deal-level and time-series evidence (confirmed).** Aventis's 64-deal cybersecurity database 2015–2025; Cogent's eight-quarter EV/Revenue and EV/EBITDA series; per-company multiples with margin and Rule-of-40 side by side.
3. **Operator-ready regulatory material.** Day-level effective dates, three-tier penalty schedules mapped to article groups, and use-case classification matrices — liftable into an exhibit with attribution intact.

4. **Breadth of advisor commentary.** Law-firm analyses (Jones Day, Linklaters, Cooley) alongside specialist explainers; note that vendor-authored material (Luware, Zango.ai) is mixed in.
5. **Methodological transparency in technical sources.** Preprints and reproducible benchmark suites stating hardware, dataset and recall/precision thresholds. *Preprints, not journal peer review.*

Contested, and not counted as advantages

- **Recency.** See What changed from v1. Not established.
- **Source authority.** Small edge at best (Jev +0.23, grok +0.05).

Default (exa) Advantages

1. **Regulator Coverage** — BaFin, Eurofi, European Commission, national supervisors, KPMG/Forvis Mazars
2. **Practitioner Perspective** — augustyniak.ai production benchmarks, vendor official benchmarks (Qdrant)
3. **Consistency** — Equally authoritative for well-established topics (public SaaS comps, major regulatory frameworks)
4. **Zero-Key Access** — Works without API key via keyless tier

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Recommendations

Consulting queries (BI / M&A / Finance / IT / Regulatory) — prefer Keenable

On the pre-declared rule (gap ≥ 0.10 and the same winner in both presentation orders) the record is **Jev 3 of 5 clear wins, frontier judge 5 of 5**. Jev's consulting interval (−0.041 to +0.430) includes zero, so the direction is consistent but the magnitude is not established. The gain is concentrated where segmented data matters:

- M&A and valuation work needing multiples split by growth, revenue or EBITDA tier
- Competitive intelligence that depends on deal-level private-market data
- Regulatory tracking that needs effective dates and penalty schedules, not summaries
- Technical due diligence where reproducible benchmarks beat vendor claims

Multi-hop deep research — do not standardise

+5.5% of scale, mixed winners. **Exa won DEEP-001 outright** (CSRD supply-chain reporting); Keenable won DEEP-002 decisively; three were splits. Pick per question, or run both.

Either engine is adequate

- Public market comparables (Bessemer, Meritech, SaaS Capital)
- Major regulatory frameworks (EU AI Act, GDPR, DORA) at overview level
- Historical trend work and vendor benchmark round-ups

Implementation Note

The default backend in this Hermes configuration falls back to **exa** (keyless tier) when ddgs is unavailable. Results shown for "Default" are exa-backed. A configured Brave/Tavily/Perplexity backend may yield different comparative results.

Appendix: Test Queries Run

The 30 queries are reproduced byte-identically, because a re-worded query is a different experiment. Canonical machine-readable source: `config/queries.yaml` in the repository. The abbreviated list that appeared in v1 has been removed.

ID	CATEGORY	LIMIT	QUERY (BYTE-IDENTICAL TO WHAT WAS EXECUTED)
BI-001	business_intelligence	10	`Total Addressable Market B2B SaaS DACH region 2024 2025 forecast`
BI-002	business_intelligence	10	`Competitive landscape European neobanks 2024 market share revenue N26 Revolut Bunq Monzo`
BI-003	business_intelligence	10	`Enterprise software spending growth trends 2024 Gartner IDC worldwide IT spending forecast`
BI-004	business_intelligence	10	`Supply chain management software market size 2024 key vendors SAP Oracle Blue Yonder Manhattan Associates`
BI-005	business_intelligence	10	`Generative AI enterprise adoption rates 2024 survey McKinsey BCG Bain Deloitte PwC`
STRAT-001	strategy_ma	10	`Technology M&A valuation multiples cybersecurity 2023 2024 EV/Revenue EV/EBITDA`
STRAT-002	strategy_ma	10	`Microsoft Activision Blizzard deal structure regulatory approval timeline FTC CMA EU Commission`
STRAT-003	strategy_ma	10	`Private equity portfolio value creation levers 2024 operational improvement add-on acquisitions`
STRAT-004	strategy_ma	10	`Carve-out separation costs benchmarks IT systems TSAs duration 2023 2024`
STRAT-005	strategy_ma	10	`Roll-up strategy success factors fragmentation HVAC dental veterinary 2024 case studies`
FIN-001	finance_markets	10	`SaaS valuation benchmarks Series B ARR multiple 2024 public comps private transactions`
FIN-002	finance_markets	10	`Private credit direct lending returns vs broadly syndicated loans 2023 2024 IRR MOIC`
FIN-003	finance_markets	10	`Venture capital fund performance quartiles vintage 2020 2021 2022 TVPI DPI RVPI`
FIN-004	finance_markets	10	`Infrastructure fund returns renewable energy digital infrastructure 2023 2024 benchmarks`
FIN-005	finance_markets	10	`SPAC performance post-merger 2021 2022 2023 redemption rates shareholder returns`
IT-001	it_technical	10	`Kubernetes adoption enterprise 2024 Gartner CNCF survey production clusters percentage`
IT-002	it_technical	10	`Vector database comparison Pinecone Weaviate Chroma Milvus Qdrant performance benchmarks 2024`
IT-003	it_technical	10	`LLM inference cost optimization GPU utilization vLLM TensorRT-LLM TGI 2024 benchmarks`
IT-004	it_technical	10	`Data mesh architecture implementation patterns Data Contracts governance 2024 case studies`
IT-005	it_technical	10	`Observability platform market share Datadog New Relic Dynatrace Grafana Splunk 2024 revenue`

ID	CATEGORY	LIMIT	QUERY (BYTE-IDENTICAL TO WHAT WAS EXECUTED)
REG-001	regulatory	10	`EU AI Act implementation timeline financial services high-risk AI systems 2024 2025`
REG-002	regulatory	10	`CSRD ESRS reporting requirements non-EU companies scope 2024 2025 double materiality`
REG-003	regulatory	10	`SOX 404 compliance cost benchmarks mid-market companies 2024 external auditor fees internal resources`
REG-004	regulatory	10	`DORA Digital Operational Resilience Act financial entities ICT risk management requirements 2025`
REG-005	regulatory	10	`GDPR Article 28 DPA requirements international data transfers SCCs 2024 Schrems III implications`
DEEP-001	deep_research	10	`Impact of CSRD on non-EU companies supply chain reporting Scope 3 emissions data availability challenges 2024`
DEEP-002	deep_research	10	`Generative AI ROI enterprise deployments 2023 2024 measurable productivity gains cost savings case studies`
DEEP-003	deep_research	10	`Climate transition risk banking portfolio alignment Paris Agreement NZBA PCAF methodology 2024`
DEEP-004	deep_research	10	`Quantum computing threat timeline RSA ECC cryptography migration post-quantum NIST standards 2024 enterprise readiness`
DEEP-005	deep_research	10	`Green hydrogen cost parity forecasts 2030 2040 electrolyzer CAPEX renewable LCOE regional analysis Europe Middle East Australia`

Scored subset. Only **10 of these 30** went through the two independent judges (Appendices A and B): BI-001, STRAT-001, REG-001, IT-002, FIN-001 (one exemplar from each consulting category) plus DEEP-001..005. The other 20 were captured but **not scored**, so nothing in this report supports a claim about them.

Protocol deviation. FIN-001 was captured at `limit=5`; every other query used `limit=10`.

Appendix A — Decision-Plane (Jev / System One) Cross-Check

> **What this appendix is.** The body of this report was scored by a single LLM judge — DeepSeek-V4.1-Flash, **unblinded**, reading the raw result sets rather than the anonymised cards used by the two appendices. It is the weakest of the three layers in evidential terms and is reported as narrative, not as measurement. This appendix adds a **second, independent scoring layer** — the VWOOSH decision plane, running TypeSafe Jev Cloud (`jev-1.13.0`) over two purpose-written MCDA contracts — and reports where the two layers agree and, more usefully, where they do not. Nothing here replaces a body finding; the decision plane either corroborates it or puts a caveat next to it — and where it disagreed, the body above has been changed rather than merely annotated.

A.1 Why a second layer at all

A single-judge narrative has two known failure modes for this kind of work:

1. **Over-claiming.** A judge that has already formed a thesis writes sweeping criteria deltas to support it. The recency gap claimed in the body (Section "Quantitative Score Summary", +0.8 of 5) is exactly this shape — and, as shown in A.5, the decision plane does *not* reproduce it.

2. **Non-reproducibility.** Prose scores cannot be re-run. A typed contract plus a pinned model plus a rubric hash can be.

The decision plane is a *different kind of judge*: ordinal anchors, a fixed rubric compiled to a typed contract, per-dimension probability distributions over the anchor set, and an explicit confidence floor that refuses to let a low-confidence call pass silently as a decision.

A.2 Execution provenance (verified, not assumed)

The `vwoosh-decision-plane` skill documents a silent-degradation trap: if `typesafe-sdk` is missing or `TYPESAFE_API_KEY` is unset, the service falls back to a local LM Studio model and inflates latency while making **zero** cloud calls. Every call in this appendix was therefore verified against the returned verdict object rather than trusted from the CLI banner.

PROPERTY	OBSERVED VALUE
Backend (all 20 evaluation + 40 arbitration calls)	`typesafe_cloud`
Model resolved	`jev-1.13.0`
Degradation flag raised	never (`degraded: false` on all 20 evaluations)
Live HTTP dispatch	`POST https://api.typesafe.ai/v1/systemone → 200` (e.g. `req_01a1ca3660387c57ba47e1fc6c5d0552`)
MCDA rubric hash	`sha256:c6fcbc1048637221244ad5ea9333a306fb4e46b469226316bc4217c0c7c0b6ce`
Arbitration rubric hash	`sha256:61fa2cd2beb2d54b660ca0452c975c6abc750218fca29b44a1da9dfcf0793029`
`evaluate` latency	n=20, min 198 ms / median 220 ms / max 368 ms
`compare` inner-call latency	n=40, median 205 ms / max 269 ms
Escalations against the 0.70 floor	10 of 20 evaluations

Calls made, counted from the artefact rather than from the plan: **20** `evaluate` (10 query pairs × 2 engines) + **20** `compare` (10 pairs × 2 presentation orders, each producing 2 verdicts) = **40 CLI invocations, 60 System One calls**. An earlier draft carried the five-category arithmetic (10 + 10 = 30 calls); that was the first pass and is superseded.

> **Cost of regeneration.** Jev is priced per call, not per token. Counted from the artefacts:

STAGE	CALLS	OBSERVED LATENCY	BASIS	EST. COST
Capture (2 engines × 30 queries)	60 HTTP	~0.5-2 s each	vendor list price	Keenable anonymous tier free; your Exa plan
Layer 2 — Jev `evaluate`	20	median 220 ms	~$0.042 / 1k actions	~$0.0008
Layer 2 — Jev `compare`	40	median 205 ms	~$0.042 / 1k actions	~$0.0017
Layer 3 — frontier judge	20	~15-120 s each	your model's per-call price	model-dependent
Layer 2 total	**60**	sub-second	—	**~$0.0025**

> Jev figures are list-price estimates, not an invoice. The dominant costs are the frontier judge and operator time; the System One layer is effectively free at this scale.

A.3 The two contracts

Both contracts live in `evaluation/contracts/` and are reproduced here in summary. Note the weight vector deliberately mirrors the human rubric, so a divergence between layers is *not* an artefact of different weighting.

`search-engine-mcda-v1` — per-engine scoring

Ordinal anchors 0..2 per dimension; composite is the weight-normalized mean on the 0..2 scale. `relevance_precision` is declared a **critical dimension** (a critical failure forces REJECT/ESCALATE rather than averaging away).

DIMENSION	WEIGHT	MIRRORS BODY CRITERION (WEIGHT)
`relevance_precision`	2.5	Relevance & Precision (25%)
`source_authority`	2.0	Source Quality & Authority (20%)
`analytical_depth`	2.0	Depth & Completeness (20%)
`recency_freshness`	1.5	Recency & Freshness (15%)
`structured_extractability`	1.0	Structured Data Extraction (10%)
`query_understanding`	1.0	Query Understanding (10%)

`search-engine-arbitration-v1` — head-to-head IDPS arbitration

Companion contract for `vwoosh decision compare`, which resolves two candidates with zero order-bias (IDPS). Dimensions are framed as *comparative* rather than absolute: `evidence_sufficiency` (3.0, critical), `dispositive_recency` (2.5), `decision_granularity` (2.5), `source_authority` (2.0), `deliverable_readiness` (2.0). `tie_epsilon: 0.03`.

Evidence cards

Because System One models have a hard input ceiling (~12,000 chars for Jev) and do not perform arithmetic, each engine's result set was compiled into a bounded evidence card containing *only* what the engine actually returned — every domain, every figure, and a **pre-computed facts block** (counts of current-vs-stale results, tables present, tier-1 entities cited) calculated by script, not estimated by the model. Cards are in `evaluation/evidence/`, named `{query_id}_{A|B}.txt`; A=Keenable, B=default. The engine identity is **not** stated inside the cards — the judge scores blind.

Scope: the decision plane was run on **all six categories, 10 query pairs** — the five consulting categories (BI-001, STRAT-001, REG-001, IT-002, FIN-001) and the five deep-research queries (DEEP-001..005), 20 evaluations and 20 order-swapped arbitrations. (The deep-research pairs were added after the first drafting pass, which covered only the five consulting categories; every figure in this appendix reflects the full ten pairs.)

A.4 Layer 1 result — MCDA scores (0–2, higher is better)

Rebuilt from the single canonical artefact `evaluation/jev/summary.json` (20 evaluations, 20 arbitrations). Figures are quoted to four decimals so that the printed Δ equals the printed difference of the two columns; an earlier draft rounded to three and the Δ column then disagreed with its own cells, and carried the five-category run besides.

QUERY	KEENABLE	DEFAULT (EXA)	Δ	MEAN CONFIDENCE K / D	IDPS WINNER
BI-001 (market sizing)	1.6035	1.0775	+0.5260	0.68 / 0.72	keenable
STRAT-001 (cyber M&A)	1.7740	1.6920	+0.0820 †	0.58 / 0.69	keenable
REG-001 (EU AI Act)	1.8390	1.6685	+0.1705	0.77 / 0.74	keenable
IT-002 (vector DB)	1.4720	1.3405	+0.1315	0.66 / 0.77	keenable
FIN-001 (SaaS multiples)	1.5125	1.4485	+0.0640 †	0.59 / 0.60	keenable
DEEP-001 (CSRD supply chain)	1.3370	1.3880	-0.0510 †	0.69 / 0.71	default-exa
DEEP-002 (GenAI ROI)	1.5540	1.0285	+0.5255	0.79 / 0.84	keenable
DEEP-003 (climate/banking)	1.5480	1.4665	+0.0815 †	0.64 / 0.82	keenable
DEEP-004 (post-quantum)	1.4255	1.4730	-0.0475 †	0.74 / 0.79	keenable
DEEP-005 (green hydrogen)	1.6235	1.5805	+0.0430 †	0.73 / 0.70	keenable

† gap below the 0.10 noise floor — **not a win**. ⚠ the two Jev contracts disagree on the winner (only DEEP-004).

Ten-pair means: Keenable 1.5689 vs default 1.4163, gap **0.1525** — on the 0–100 presentation scale, 78.4 vs 70.8. See the statistical cautions in "Quantitative Score Summary" before quoting that as a percentage.

A.5 Layer 2 result — IDPS arbitration (blind, both orders)

Each pair was arbitrated twice with the candidates swapped. A winner that flips between orders indicates position bias rather than a real difference. Values read from `evaluation/jev/summary.json`; an earlier draft of this table carried the five-category run.

QUERY	ORDER	SCORE A	SCORE B	Δ	WINNER	MAPS TO
BI-001	A=kee / B=def	1.258	0.723	+0.536	candidate_a	**Keenable**
BI-001	A=def / B=kee	0.720	1.244	-0.524	candidate_b	**Keenable**
STRAT-001	A=kee / B=def	1.734	1.437	+0.298	candidate_a	**Keenable**
STRAT-001	A=def / B=kee	1.427	1.728	-0.301	candidate_b	**Keenable**
REG-001	A=kee / B=def	1.857	1.504	+0.353	candidate_a	**Keenable**
REG-001	A=def / B=kee	1.501	1.886	-0.385	candidate_b	**Keenable**
IT-002	A=kee / B=def	1.341	1.045	+0.296	candidate_a	**Keenable**
IT-002	A=def / B=kee	1.052	1.330	-0.277	candidate_b	**Keenable**
FIN-001	A=kee / B=def	1.533	1.436	+0.097	candidate_a	**Keenable**
FIN-001	A=def / B=kee	1.428	1.526	-0.098	candidate_b	**Keenable**
DEEP-001	A=kee / B=def	1.117	1.212	-0.095	candidate_b	**default**
DEEP-001	A=def / B=kee	1.200	1.124	+0.076	candidate_a	**default**
DEEP-002	A=kee / B=def	1.331	0.710	+0.621	candidate_a	**Keenable**
DEEP-002	A=def / B=kee	0.723	1.327	-0.604	candidate_b	**Keenable**
DEEP-003	A=kee / B=def	1.274	1.201	+0.073	candidate_a	**Keenable**
DEEP-003	A=def / B=kee	1.206	1.262	-0.055	candidate_b	**Keenable**
DEEP-004	A=kee / B=def	1.366	1.190	+0.176	candidate_a	**Keenable**
DEEP-004	A=def / B=kee	1.195	1.379	-0.185	candidate_b	**Keenable**
DEEP-005	A=kee / B=def	1.333	1.272	+0.060	candidate_a	**Keenable**
DEEP-005	A=def / B=kee	1.265	1.322	-0.056	candidate_b	**Keenable**

Order-invariance of the winner: PASS, 10/10. No pair produced a tie (`tie_broken_by` null throughout), so the `tie_epsilon: 0.03` mechanism never fired.

Arbitration dimension means over all 20 verdicts:

DIMENSION	KEENABLE	DEFAULT	Δ
`source_authority`	1.621	1.319	**+0.302**
`decision_granularity`	1.524	1.227	**+0.297**
`deliverable_readiness`	0.979	0.739	**+0.240**
`evidence_sufficiency`	1.274	1.076	**+0.198**
`dispositive_recency`	1.652	1.463	**+0.189**

A.6 How the decision plane changed the findings

(1) It removed the recency over-claim

The body's v1 draft asserted a **+0.8 of 5** recency advantage. Neither Jev contract supports an effect of that size:

LAYER	RECENCY GAP	AS % OF ITS SCALE	N
Body v1 (single unblinded judge)	+0.8 of 5	16.0%	6 exemplars
MCDA contract, all 10 pairs	**+0.042 of 2**	**2.1%**	20 evaluations
MCDA contract, consulting 5	+0.046 of 2	2.3%	10 evaluations
Arbitration (`dispositive_recency`)	+0.280 of 2	14.0%	20 verdicts

The two Jev contracts disagree with each other *and* with the body — a finding in its own right. Per-query MCDA recency is not uniform: Keenable leads BI-001 (**1.40 vs 1.11**), **trails** FIN-001 (**1.28 vs 1.56**), and is essentially level on STRAT-001 (**1.49 vs 1.47**). Recency is contested and small in aggregate; the v1 +0.8 claim was an overstatement.

(2) It re-ranked *which* dimension drives the difference

DIMENSION	Δ ALL 10	Δ CONSULTING 5
`analytical_depth`	**+0.308**	**+0.424**
`source_authority`	+0.232	+0.144
`query_understanding`	+0.168	+0.252
`structured_extractability`	+0.122	+0.276
`relevance_precision`	+0.037	+0.086
`recency_freshness`	**+0.042**	+0.046

The advantage is in disaggregation, not currency. **Buy Keenable for resolution, not recency.**

(3) Calibration signal

Escalations against the 0.70 confidence floor: **10 of 20** evaluations across all pairs (7 of 10 on the consulting five). `recency_freshness` carries the least confidence — mean **0.483** over all 20, with **15 of 20** below the floor (consulting-only: 0.414, 8 of 10). Next lowest is `query_understanding` (0.592 all-20). The plane is confident about what it can count and unsure about what requires judging currency or intent.

(4) Order bias: winner stable, magnitude not

QUERY	|Δ| AB	|Δ| BA	ASYMMETRY	AS % OF Δ	VS `TIE_EPSILON` 0.03
BI-001	0.536	0.524	0.012	2.3%	within
STRAT-001	0.298	0.301	0.003	1.1%	within
REG-001	0.353	0.385	0.032	8.6%	**exceeds**
IT-002	0.296	0.277	0.019	6.5%	within
FIN-001	0.097	0.098	0.000	0.4%	within
DEEP-001	0.095	0.076	0.018	21.5%	within
DEEP-002	0.621	0.604	0.017	2.8%	within
DEEP-003	0.073	0.055	0.017	26.7%	within
DEEP-004	0.176	0.185	0.009	4.9%	within
DEEP-005	0.060	0.056	0.004	7.1%	within

Mean asymmetry **8.2%** of delta magnitude, maximum **26.7%**. Only REG-001 exceeds the contract's 0.03 absolute `tie_epsilon`. Consequences:

- Rankings from `compare` are trustworthy (**10/10** stable).
- Delta magnitudes should not be quoted to two decimals; for |Δ| < 0.15 the magnitude is not reliable at all.

A.7 Agreement between the layers

QUERY	PLANE Δ (MCDA, K–D)	PLANE WINNER (IDPS)	BODY VERDICT (1–5)	AGREE?
BI-001	+0.526	Keenable	Keenable (+0.5)	yes
STRAT-001	+0.082	Keenable	Keenable (+0.6)	yes
REG-001	+0.171	Keenable	Keenable (+0.3)	yes
IT-002	+0.131	Keenable	Keenable (+0.2)	yes
FIN-001	+0.064	Keenable	tie (±0.0)	**no**

4 of 5 in direction; the single disagreement (FIN-001) is about size, not sign.

Do not compare the layers' absolute numbers. Different scales, different anchors:

LAYER	KEENABLE	DEFAULT	GAP
Body, weighted (1–5) — **withdrawn**, see Quantitative Score Summary	~~4.56~~	~~4.28~~	~~0.28~~
MCDA composite (0–2), all 10 pairs	1.569	1.416	0.153
MCDA composite (0–100 presentation)	78.4	70.8	7.6
MCDA rescaled to 1–5 (`1 + 2 × composite`)	4.14	3.83	0.31

The 1–5 rescaling is `1 + 2 × composite` (mapping 0–2 onto 1–5). An earlier draft used `1 + composite`, which maps onto 1–3 and was wrong.

A.8 Reproducing this appendix

```
cd ~/Projects/search-benchmark

# 1. Confirm the cloud backend is live (guards against silent local fallback).
vwoosh decision evaluate \
  --state-file evaluation/jev_probe_state.json \
  --contract ~/.hermes/vwoosh-local/engine/_DEVELOPMENT/00_ADMIN/decisions/dependency-selector-v1.yaml \
  --output evaluation/jev_probe_verdict.json --backend cloud
# expect: Backend: typesafe_cloud / Model: jev-1.13.0

# 2. Re-run both layers (10 evaluate + 10 compare, both orders).
python3 scripts/run_decision_plane.py

# 3. Artefacts
#    evaluation/jev/mcda_scores.csv     per-engine per-dimension scores
#    evaluation/jev/arbitration.csv     both orders, deltas, winners
#    evaluation/jev/summary.json        full raw verdicts incl. probabilities
#    evaluation/jev/mcda_<q>_<A|B>.json individual verdicts
#    evaluation/jev/arb_<q>_<AB|BA>.json individual arbitrations
```

Operational notes worth carrying forward:

- `--backend cloud` must be passed explicitly; `auto` may select local.
- Verify `verdict.backend == "typesafe_cloud"` in the **output JSON**, not the banner — the silent-fallback failure mode produces a normal-looking table with zero cloud calls.
- Jev's input ceiling forces summarising before evaluation. Every number in the evidence cards was computed by script first and injected as text; System One does not do arithmetic, so feeding it raw prose and asking for a rate would have produced invented figures.
- Blind labelling is enforced by withholding engine identity inside the cards. Re-running `compare` in both orders is the cheap way to check the blind held.

A.9 Net effect on the report's conclusions

BODY CONCLUSION	AFTER DECISION-PLANE CROSS-CHECK
Keenable wins overall	**Upheld, with the margin restated. 10/10** arbitration wins, both orders, no ties; mean MCDA gap **+0.153** over all 10 pairs (+0.195 on the consulting five). The v1 wording ("5/5 wins, +0.201") was the five-category run.
Recency is the largest differentiator	**Revised.** Likely overstated; recency is category-specific and small in aggregate.
Depth/granularity is an advantage	**Reinforced.** Largest gap in both contracts.
Source authority is a tie	**Not upheld.** MCDA all-10 puts Keenable ahead by **+0.232** (consulting five: +0.144); arbitration agrees (+0.223). Treat as a real but modest edge, not a tie.
FIN-001 is a tie	**Upheld in substance.** Plane's +0.069 is within noise.
Default wins on regulator coverage	**Not corroborated.** `source_authority` favours Keenable in both contracts. The body's regulator-coverage point stands as a *qualitative* observation but is not reflected in the blind scores.

Bottom line of this appendix: the primary recommendation survives an independent judge, but the *reasoning* given for it in the body was partly wrong. Keenable's advantage is in resolution and deliverable-readiness, not primarily in freshness — and the body's own confidence in the recency claim was too high.

Appendix B — Third Layer: Frontier-LLM Judge (grok-4.6), All Ten Pairs

B.1 Purpose and design

Appendix A added one independent judge (Jev) against the single-judge body. Two judges can only disagree; three can reveal *which* judge is the outlier on *which* dimension. This layer adds a frontier LLM judge — the tool a consultant would actually reach for — under the same discipline as the IDPS layer: **blind cards, both presentation orders.**

PROPERTY	VALUE
Model	`grok-4.6` via `custom:vwoosh-foundry` (Azure Foundry, credit-backed)
Dimensions	the **same six** and **same ordinal anchors (0..2)** as `search-engine-mcda-v1`
Weights	identical (2.5 / 2.0 / 2.0 / 1.5 / 1.0 / 1.0) — composites directly comparable to Jev MCDA
Blindness	engine identity withheld inside cards; both presentation orders run
Calls	**20** (10 query pairs × 2 orders)
Validation	strict-JSON output, integer 0..2 per dimension per candidate, winner ∈ {A,B,tie}, up to 3 retries

B.2 Three layers, all ten query pairs

K = Keenable · D = default (exa) · `–` means the judge called it a tie.

QUERY	BODY (1–5)	JEV MCDA (0–2)	JEV IDPS	GROK-4.6 (0–2)	CONSENSUS
BI-001	4.5 / 4.0 → K	1.603 / 1.077 → K	K	1.350 / 0.925 → K	**all agree**
STRAT-001	4.8 / 4.2 → K	1.774 / 1.692 → K	K	1.900 / 1.550 → K	**all agree**
REG-001	4.6 / 4.3 → K	1.839 / 1.668 → K	K	1.800 / 1.500 → K	**all agree**
IT-002	4.7 / 4.5 → K	1.472 / 1.341 → K	K	1.800 / 1.400 → K	**all agree**
FIN-001	4.3 / 4.3 → tie	1.512 / 1.449 → K	K	1.750 / 1.250 → K	3 of 4 say K
DEEP-001	*aggregate only*	1.337 / **1.388 → D**	**D**	1.450 / 1.450 → tie	**default or no difference**
DEEP-002	*aggregate only*	1.554 / 1.028 → K	K	1.800 / 1.300 → K	**all agree: K**
DEEP-003	*aggregate only*	1.548 / 1.466 → K	K	1.500 / **1.550 → D**	split, 2–1 K
DEEP-004	*aggregate only*	1.425 / **1.473 → D**	K	1.750 / 1.450 → K	split, 2–1 K
DEEP-005	*aggregate only*	1.623 / 1.581 → K	K	1.300 / **1.400 → D**	split, 2–1 K

Tally: **Jev IDPS** Keenable 9/10 · **Jev MCDA** Keenable 8/10 · **grok-4.6** Keenable 7/10, default 2/10, 1 tie.

B.3 The headline result: the advantage is category-specific

SLICE	JEV MCDA Δ	AS % OF SCALE	GROK Δ	AS % OF SCALE	FRONTIER/JEV RATIO
All 10 pairs	+0.153	7.6%	+0.262	13.1%	1.7× †
Consulting 5 (BI, STRAT, REG, IT, FIN)	+0.195	9.7%	+0.395	19.8%	**2.0×** †
Deep research 5 (DEEP-001..005)	+0.110	5.5%	+0.130	6.5%	**1.2×** †

† Ratios of mean ordinal composites, **not effect sizes**. The scales are not interval and the confidence intervals in the Verdict overlap. Read the column as "the frontier judge is more generous to the challenger", not as a measured multiplier: with 95% intervals of −0.041 to +0.430 (Jev consulting) and −0.187 to +0.407 (Jev deep), the two means cannot be separated at n=5.

The direction is consistent across slices, but **the v1 claim that the consulting edge is twice the deep-research edge is not supported and is withdrawn**. In deep research the three judges agree on only one of five queries:

- **DEEP-002 (GenAI ROI) — clear Keenable win.** Unanimous, and the largest gap in the whole benchmark (Jev 1.554 vs 1.028). The evidence cards explain it: the default engine returned five distinct domains with three Microsoft and three Automation Anywhere vendor case studies, while the challenger returned nine distinct domains spanning arXiv, Omdia, McKinsey, Deloitte and independent research.
- **DEEP-001 (CSRD supply chain) — directional default edge, below the clear-win floor.** The only query where two of three judges favour the incumbent, and the only IDPS loss for Keenable (Δ −0.095).
- **DEEP-003, DEEP-004, DEEP-005 — contested 2–1 for Keenable in every case**, but the dissent is always the frontier judge, and in two of the three the frontier judge's own verdict flipped between presentation orders.

This materially qualifies the report body's Deep Research row, which recorded both engines at 4.2/4.2. That aggregate concealed a genuine win and a genuine loss.

B.4 The frontier judge polarises more — but the gap depends on the category

All 10 pairs: Jev Δ = +0.153 (7.6% of scale); grok Δ = +0.262 (13.1%). The frontier judge's mean delta is larger in every slice — by a factor of 1.7 across all pairs, 2.0 on consulting and 1.2 on deep research. **Those factors are ratios of ordinal composites, not effect sizes**, and the underlying intervals overlap; they indicate direction and rough magnitude only (see the caveat in §B.3). Prior conclusion refined: the frontier judge does not simply inflate everything; it inflates **structured, benchmark-style questions** where there are clear anchors to distinguish, and behaves more like System One on open-ended synthesis questions.

B.5 Position stability — and a judge that contradicts itself

Two definitions of "winner" are available here, and they differ. The judge returns a declared `winner` field *and* six per-dimension scores from which a weighted composite is derived. Both are reported:

DEFINITION	WINNER STABLE	FLIPS
Declared `winner` field	**8/10**	DEEP-001, DEEP-003
Weighted composite	**7/10**	DEEP-001, DEEP-003, DEEP-005

Score drift between orders: grok 0.110 (K) / 0.165 (D), max **0.700**; Jev ≈0.01–0.03, max 0.032.

Under the declared-winner definition, both flips moved from default winning to Keenable winning, i.e. the **candidate in the second position won in each ordering**:

QUERY	ORDER A=K, B=D	ORDER A=D, B=K	DECLARED WINNER
DEEP-001	**D** wins (1.400 / 1.800)	**K** wins (1.500 / 1.100)	position B
DEEP-003	**D** wins (1.400 / 1.650)	**K** wins (1.600 / 1.450)	position B

Two flips is thin evidence, so this is stated as a **suggestive position-B tendency, not an established bias**: both observed flips favoured the second-shown candidate, and both occurred where the true difference is small (Jev |Δ| = 0.085 and 0.064, the two smallest in the DEEP set). On DEEP-001 the incumbent's score moved **0.700** between orders (1.800 → 1.100) while the challenger moved 0.100 — an asymmetry far larger than any effect being measured.

DEEP-005 is a separate and more useful defect. The judge scored Keenable 1.400 vs default 1.250 — a composite margin of +0.15 in the challenger's favour — yet **declared the default engine the winner**. The stated verdict contradicts the judge's own scores. This is a reminder that a frontier judge's headline verdict is a holistic judgement, not a derivation from its rubric outputs, and the two must be cross-checked before either is quoted.

Operational consequence: a single-pass frontier judgement of a close comparison is unreliable. Winner stability (8/10 declared, 7/10 by composite) is high enough to trust the direction on clear-cut questions, but the 0.700 worst-case score drift means the magnitudes, and any verdict on a near-tie, should not be quoted without both presentation orders — and should be cross-checked against the IDPS layer, which was stable 10/10 on the same pairs.

B.6 Where the layers agree and disagree, by dimension

Mean score per dimension over all ten pairs (Keenable / default):

DIMENSION	GROK Δ	JEV Δ	READING
`analytical_depth`	**+0.50**	+0.31	**both: Keenable** — the durable finding
`query_understanding`	+0.45	+0.17	both: Keenable
`recency_freshness`	**+0.60**	**+0.04**	**irreconcilable**
`structured_extractability`	+0.05	+0.12	both: near-parity on the full set
`source_authority`	+0.05	+0.23	both: small Keenable edge
`relevance_precision`	+0.05	+0.04	both: parity

Two things stand out:

1. **`analytical_depth` is the only dimension where both judges find a large, same-signed effect.** This is the defensible core: the challenger disaggregates where the incumbent generalises. Every layer agrees.
2. **`recency_freshness` remains irreconcilable** — grok +0.60, Jev +0.04. And `recency_freshness` is Jev's lowest-confidence dimension across all 20 evaluations (mean confidence **0.483**; 10 of 20 evaluations escalated below the 0.70 floor). The dimension the judges fight over is precisely the one System One says it cannot read confidently from summarised evidence.

Note also that `structured_extractability` collapses to near-parity (+0.05) once deep-research queries are included, whereas on the consulting five alone it was +0.30. The Keenable "lift-ready tables" advantage is concentrated in the consulting categories, not universal.

B.7 Net effect on the report's conclusions

BODY CONCLUSION	AFTER ALL THREE LAYERS
Keenable wins overall	**Upheld**, but mostly by the consulting categories (9/10 IDPS wins)
Advantage is large and general	**Revised.** Consulting: 9.7% of scale. Deep research: 5.5%, with 1 win, 1 loss, 3 splits
Recency is the biggest differentiator	**Rejected.** Both judges put recency at or near zero effect in aggregate (Jev +0.04)
Depth/granularity is the advantage	**Reinforced and promoted to the headline finding** (+0.31 Jev / +0.50 grok)
Deep Research is a 4.2 / 4.2 tie	**Rejected.** It concealed a clear win (DEEP-002) and a clear loss (DEEP-001)
Source authority is a tie	**Narrowed to "small Keenable edge"** (Jev +0.23, grok +0.05)

B.8 Limitations

- **n=10 pairs, but NOT 5 per category.** Five pairs are *single exemplars* from five different consulting categories, plus five deep-research queries. There is no within-category replication anywhere except deep research: a "category" label on a consulting row means one query.
- Differences below ~0.10 on the 0–2 scale are treated as no difference. At that floor the Jev consulting tally is 3/5, not 5/5.
- **The layers are not independent instruments.** Layer 1 (body) scored *raw result sets unblinded*, and its model is unnamed. Layers 2 and 3 scored *blind evidence cards*. So L2 and L3 share a summarisation step (a card-writing bias affects both identically and is invisible here), while L1 and L3 may be the same model family. Agreement is reported as agreement, not as validation by independent instruments.
- One frontier model only. The polarisation multiple and the two observed position flips are measured facts about `grok-4.6`, not about frontier judges in general. Two flips is n=2.
- Prompt sensitivity, temperature effects and rubric-anchor wording were not ablated.
- Query strings: consulting queries ran at `limit=10` **except FIN-001 at** `limit=5` (a protocol deviation; excluded from slice means); DEEP queries all at `limit=10`.

B.9 Reproducing

```
cd ~/Projects/search-benchmark
python3 scripts/run_judge_layer.py      # 20 calls, cached/resumable, parallel-safe
python3 scripts/three_way.py            # consolidated three-way + agreement + stability
# artefacts: evaluation/judge/judge_<q>_<AB|BA>.json, evaluation/judge/summary.json,
#            evaluation/three_way.json
```